

高速采集仪表

High speed acquisition instrument

使用说明书

User manual

V3.00

安全提示 Safety Tips

1. 本产品具有独特的抗干扰设计和滤波算法，请务必可靠接地，且与交流接地线分开。
 2. 确保在符合本产品技术指标的环境中使用。
 3. 避免阳光直射。
 4. 应由具有相应专业知识并能安全操作的人员负责本产品的安装配线与维护工作。
 5. 严禁带电打开机箱。
1. This product has a unique anti-interference design and filtering algorithm. Please ensure reliable grounding and separate it from the AC grounding wire.
 2. Ensure use in an environment that meets the technical specifications of this product.
 3. Avoid direct sunlight.
 4. Personnel with corresponding professional knowledge and the ability to operate safely should be responsible for the installation, wiring, and maintenance of this product.
 5. It is strictly prohibited to open the chassis with electricity.

技术指标 Technical indicators

显示：双 5 位段码管显示

工作电源：DC10-28V, AC85-265V（额定 100-240V, 50/60HZ），宽电压供电，选其一

带载能力：8 只 350 Ω 传感器

信号范围：0.5~3.0mV/V

激励电压：5V

采样频率：10，40，640，1280HZ 可设置

内部分辨率：24 位 Σ - Δ ADC，内部分辨率 1/1000000

显示分辨率：1/100000

非线性度：0.005%F.S

最大功耗：5W

工作温度：-25~+45 $^{\circ}$ C

相对湿度：<85%RH

重量：约 300g

外形尺寸：96 $\times$ 48 $\times$ 110mm（控制柜开孔尺寸 92 $\times$ 46mm）

Display: Double 5-digit segment code tube display

Working power supply: DC10-28V, AC85-265V (rated 100-240V, 50/60Hz), wide voltage power supply, choose one of them

Carrying capacity: 8 350 Ω sensors

Signal range: 0.5~3.0mV/V

Excitation voltage: 5V

Sampling frequency: 10, 40, 640, 1280Hz, adjustable

Internal resolution: 24 bits Σ - Δ ADC, internal resolution 1/1000000

Display resolution: 1/100000

Non linearity: 0.005% F.S

Maximum power consumption: 5W

Working temperature: -25~+45 $^{\circ}$ C

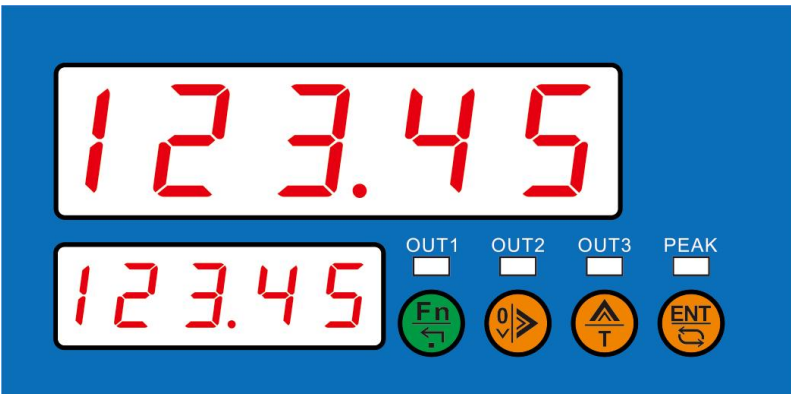
Relative humidity: < 85% RH

Weight: approximately 300g
External dimensions: 96 x 48 x 110mm (control cabinet opening size 92 x 46mm)

型号及选型 Model and selection

型号	选型功能说明
基础款	双五位显示，3 组继电器，一组开关量输入
模拟量输出款	基础款+模拟量输出（0-5V,0-10V,0-5-10V,0-20mA,4-20mA,4-12-20mA）
RS485 输出款	基础款+RS485 输出
全功能输出款	基础款+模拟量输出+RS485 输出

显示窗口及按键



- 双五位显示窗口，主屏显示净重，副屏显示设置参数、总重、峰值、单位。

Double five digit display window, with the main screen displaying net weight and the secondary screen displaying set parameters, total weight, peak value, and unit

- ：进入菜单/返回上一级，**长按进入小数点选择界面。**

Enter the menu/return to the previous level, long press to enter the decimal point selection interface

- ：进入标定界面，移动光标，主菜单界面和参数选择界面向下移动，**长按进入置零选项。**

Enter the calibration interface, move the cursor, move the main menu interface and parameter selection interface downwards, long press to enter the zero setting option

- ：改变菜单，增加数值，**长按进入去皮选项。**

Change the menu, add values, long press to enter the peeling option

- ：菜单界面为确认菜单，显示界面为切换第二窗口显示，单位，总重，峰值切换，**长按进入清除峰值界面。**

The menu interface is a confirmation menu, and the display interface is to switch to the second window display, which includes units, total weight, and peak value. Long press to enter the clear peak value interface

- PEAK 灯常亮第二窗口为峰值显示，PEAK 灯闪烁为等待触发峰值状态。

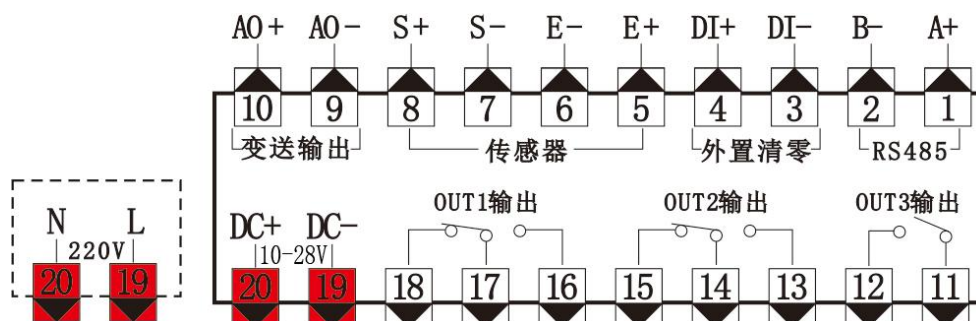
The PEAK light is constantly on, and the second window is the peak display. The PEAK light flashes to wait for the peak state to be triggered

- OUT1/OUT2/OUT3 灯亮分别为继电器 1,2,3 触发状态，熄灭为未触发状态。

Out1/Out2/Out3 lights on indicate the triggering status of relays 1, 2, and 3 respectively, while lights off indicate the non

triggering status

接口定义 Interface Definition



端子说明 Terminal Description:

- 19,20 号端子为仪表供电接口，【DC+,DC-】为 10-28V 直流电源，【N,L】为 85-265V（额定 100-240V）交流电源。Terminals 19 and 20 are the power supply interface for the instrument. 【DC+, DC -】 is a 10-28V DC power supply, and 【N, L】 is an 85-265V (rated 100-240V) AC power supply.

- 【OUT1】【OUT2】【OUT3】为继电器报警输出端子，均为无源干触点，可以设置为上限输出，下限输出，区间内输出，区间外输出，详见菜单 **C3.rEL** 继电器设置。

[OUT1] [OUT2] [OUT3] are relay alarm output terminals, all of which are passive dry contacts and can be set as upper limit output, lower limit output, and interval output

Internal output, interval external output, see menu C3.rEL relay settings for details.

- 【变送输出】（选配），可以输出 0-20mA,4-20mA,4-12-20mA,0-3.3V,0-5V,0-10V,0-5-10V 等模拟量信号，详见菜单

C6.ANA 模拟量输出。

【Transmission Output】(optional), can output analog signals such as 0-20mA, 4-20mA, 4-12-20mA, 0-3.3V, 0-5V, 0-10V, 0-5-10V, etc. Please refer to the menu for details

C6.ANA analog output.

- 【传感器】S+:传感器信号+, S -:传感器信号-, E -:传感器激励-, E+: 传感器激励+, 本仪表在压载校准模式下可忽略传感器信号正负，可以直接标定成正值。

[Sensor] S+: Sensor signal+, S -: Sensor signal -, E -: Sensor excitation -, E+: Sensor excitation+, this instrument can flicker in ballast calibration mode

The sensor signal is slightly positive or negative, and can be directly calibrated to a positive value.

- 【外置清零】将 3,4 号端子短接可实现以下功能①tPEEL 去皮（掉电不保存）②SPEEL 去皮（掉电保存）③CPEEL 取消去皮（掉电保存）④SZero 置零（掉电不保存）⑤CZero 零点校准（掉电保存）⑥REMAX 清除峰值，默认此功能关闭，详见菜单 **C1.Sys** 系统设置。

【External reset】Short circuiting terminals 3 and 4 can achieve the following functions: ① tPEEL peeling (not saved when power is off) ② SPEEL peeling (saved when power is off)

③ CPEEL cancels peeling (save after power failure) ④ SZero zeros (do not save after power failure) ⑤ CZero zero point calibration (save after power failure) ⑥ REMAX clears peak value. This function is disabled by default. Please refer to menu

C1.SyS system settings for details.

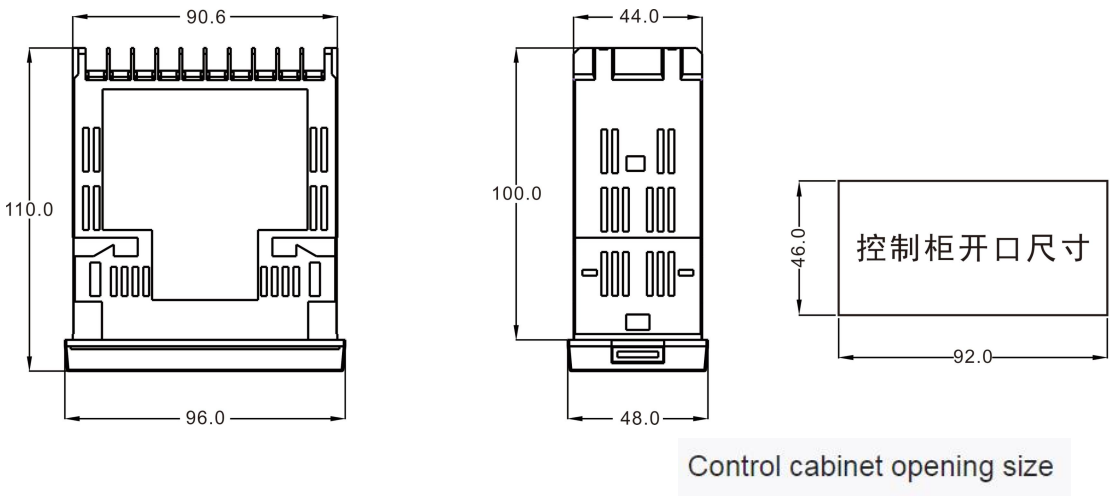
- 【RS485】（选配）MODBUS-RTU 协议，最高传输速率 1280HZ，支持主动发送，发送频率最高 1000HZ, 详见菜单

C5.CON 485 通讯设置。

RS485 (optional) MODBUS-RTU protocol, with a maximum transmission rate of 1280Hz, supports active transmission, and a maximum transmission frequency of 1000Hz. Please refer to the menu for details

C5.CON 485 communication settings.

外形及开口尺寸 Appearance and opening dimensions



功能参数一览表 List of Functional Parameters

主菜单		二级菜单				
提示符	功能	提示符	功能	默认值	范围	功能描述
C1.SyS	系统设置 System settings	100.SP	采样速率 Sampling rate	40H	10, 40, 640, 1280HZ	四种采样速率可选 Four sampling rates available
		101.GA	增益调节 Gain adjustment	128B	1, 2, 64, 128B	AD 增益调节可选 AD gain adjustment optional
		102.ME	中值滤波 median filtering	5	1, 3, 5, 9	从 x 个值中取中间值为有效值 Take the middle value from x values as the valid value
		103.rV	平均滤波 Average filtering	5	1-50	取 X 个值的平均值为有效值，设置值越大重量显示越稳定，但响应速度变慢 Take the average of X values as the valid value. The larger the set value, the more stable the weight display will be, but the response speed will

					slow down	
		104.ds	副屏默认显示 Secondary screen default display	uNit	0: Gross 1: pEak 2: uNit	0:总重 1: 峰值 2:单位 0: Total weight 1: Peak 2: Units
		105.uN	单位选择 Unit selection	kG	0-9	0: NoNE (none) 1: G (g) 2: kG (kg) 3: t (T) 4: N (N) 5: pa (pa) 6: kpa (Kpa) 7: mpa (Mpa) 8: N.M (N·m) 9: kN (kN)
		106.bi	小数点位置 Decimal point position	0	0-4	上键切换小数点位置 (循环) Up key to switch decimal point position (loop) 88888,8888.8,888.88, 88.888,8.8888
		107.dV	分度值 Graduation value	1		1, 2, 5, 10, 20, 50, 100
		108.ro	最大称量 Maximum weighing	99999	0-99999	传感器容许承载最大重量, 超过 设置值第二屏幕显示 AL.oL 最大称量≤传感器总量程-皮重 The maximum weight allowed by the sensor exceeds the set value, and the second screen displays AL.oL Maximum weighing ≤ total sensor range - tare weight
		109.di	DI 开关输入 DI switch input	NoNE	0-5	0: NoNE (invalid) 1: TPEEL (skin removed but not saved) 2: SPEEL (Peeled Storage) 3: CPEEL (Cancel Peeling Save) 4: SZero (set to zero without saving) 5: CZero (zero calibration save) 6: REMAX (clear peak)
		110.MZ	峰值判断门限 Peak judgment threshold	50	0-50000	超过设定值峰值重新检测, 设置 为 0 时, 只有当新的峰值比当前 原峰值大才重新刷新。 If the peak value exceeds the set value, it will be re detected. When set to 0, it will only be refreshed when the new peak

						value is larger than the current original peak value.	
		111.MN	峰值检测间隔 Peak detection interval	0.5	0-5.000	超过设定值且大于判断门限峰值重新检测（秒） Detect again after exceeding the set value and exceeding the judgment threshold peak value (seconds)	
		112.br	显示亮度 Display brightness	L5	L1-L8	段码管显示亮度调节 Brightness adjustment of segment code tube display	
		113.uP	显示刷新时间 Display refresh time	0.02	0.02-1.000	根据设定值刷新显示（秒） Refresh display based on set value (seconds)	
		114.bp	备份参数 Backup parameters	NO	YES/NO	备份当前的配置参数，包括校准参数，需要时可以载入当前的备份，避免恢复出厂带来的参数清除。 Backup the current configuration parameters, including calibration parameters, and load the current backup when needed to avoid parameter clearing caused by factory recovery.	仪表会重启，请断开外接被控制设备电源 The instrument will restart, please disconnect the power supply of the external controlled equipment
		115.Lp	载入备份参数 Load backup parameters	NO	YES/NO		
		116.FA	恢复出厂设置 Restore factory settings	NO	YES/NO	恢复出厂默认参数（谨慎操作）Restore factory default parameters (operate with caution)	
C2.CAL	校准参数 Calibration parameters	200.ZE	零点校准 Zero calibration		AD 内码值 AD internal code value	进入后显示 AD 值，按上键清零同时保存 After entering, display the AD value, press the up key to reset and save at the same time	
		201.Mo	校准模式 Calibration mode	Load	Load/data	Load 压载校准，data 数字校准 Load ballast calibration, data digital calibration	

			202.WE	砝码值 Weight value			压载校准时输入的实际重量值 Actual weight value input during ballast calibration	
			203.rA	传感器量程 Sensor Range		10-99999	单个传感器量程 X 传感器个数 Single sensor range X number of sensors	仅在 data 数字校准时显示 13/10000 实时翻译划译 Display only during data digital calibration
			204.SE	传感器灵敏度 Sensor sensitivity	2.000	0.010-10.000	(mv/v)	
			205.rE	传感器激励电压 Sensor excitation voltage	5	0.010-10.000	V	
			206.rV	量程修正系数 Range correction factor	1.000	0.010-2.000	通过修改该系数进一步修正传感器线性 Further correct the linearity of the sensor by modifying the coefficient	
			207.mE	多点修正使能 Multi point correction enable	0		0:CLosE （关闭） 1:opEN （打开）	
			208.mC	多点校准 Multipoint calibration			详见操作示例 Please refer to the operation example for details	
209.Mr	校准点数 Calibration points							
C3.rEL	继电器输出 Relay output	继电器 1 Relay 1	300.m1	OUT1 工作模式 OUT1 working mode	0	0-4	0: NoNE （无输出） 1: hiGh （上限输出） 2: Low （下限输出） 3: iNraN （区间内输出） 4: outrN （区间外输出） 0: NoNE (no output) 1: HiGh (upper limit output) 2: Low (lower limit output) 3: iNraN (interval output) 4: OutrN (out of range output)	
			301.V1	OUT1 数据类型 OUT1 data type	0	0-2	0: GroSS （总重） 1: NEt （净重） 2: pEak （峰值） 0: GroSS (total weight) 1: NEt (net weight) 2: PEak (peak)	

			302.r1	OUT1 回差值 OUT1 return difference	0	0.001-1	通过设置回差值避免继电器在报警点频繁动作，0.500 为报警值的 50% By setting the return difference to avoid frequent relay actions at the alarm point, 0.500 is 50% of the alarm value
			303.t1	OUT1 动作延时 Out1 action delay	0	0-9.999	继电器延时动作（秒） Relay delay action (seconds)
			304.h1	OUT1 输出上限 OUT1 output upper limit	9000	0-99999	继电器输出上限值 Relay output upper limit value
			305.L1	OUT1 输出下限 OUT1 output lower limit	1000	0-99999	继电器输出下限值 Lower limit value of relay output
		继电器 2 Relay 2	306.m2	OUT2 工作模式	0	0-4	同继电器 1 Same as relay 1
			307.V2	OUT2 数据类型	0	0-2	
			308.r2	OUT2 回差值	0	0.001-1	
			309.t2	OUT2 动作延时	0	0-9.999	
			310.h2	OUT2 输出上限		0-99999	
			311.L2	OUT2 输出下限		0-99999	
		继电器 3 Relay 3	312.m3	OUT3 工作模式	0	0-4	
			313.V3	OUT3 数据类型	0	0-2	
			314.R3	OUT3 回差值	0	0.001-1	
			315.t3	OUT3 动作延时	0	0-9.999	
			316.h3	OUT3 输出上限		0-99999	
			317.L3	OUT3 输出下限		0-99999	

C4.AdV	高级应用 Advanced Applications	400.CV	蠕变跟踪 Creep tracking	0	0-10	10 个等级，根据真实蠕变调整，0: 无效 10 levels, adjusted based on actual creep, 0: invalid
		401.dZ	显示归零范围 Display zeroing range	0	0-50000	当重量绝对值在此范围时显示归零（此参数只是将显示归零，实际重量值还存在，并且参与运算）例如：传感器实际零点为 4，当前参数设置为 5，显示会是 0，当压重为 20 时，仪表会显示 24。 When the absolute weight value is within this range, it will be displayed as zero (this parameter only shows zero, the actual weight value still exists and

						participates in the calculation). For example, if the sensor's actual zero point is 4 and the current parameter is set to 5, the display will be 0. When the weight is 20, the instrument will display 24
		402.tV	动态追踪范围 Dynamic tracking range	0	0-50000	0:不开启动态追踪，详见【名词解释】 0: Do not enable dynamic tracking, see [noun explanation] for details
		403.tC	动态追踪刷新 Dynamic tracking refresh	0.2	0-2.000	0: Do not enable dynamic tracking, see [noun explanation] for details
		404.SV	稳定重量开关 Stable weight switch	0	0-1	0: close 1: Open 开启时重量显示会直接显示最终值，例如实际重量 200，显示会从 0 直接显示 200，不显示 0-200 的变化过程。 0: close 1: When Open is turned on, the weight display will directly display the final value. For example, if the actual weight is 200, the display will start from 0 and directly display 200, without displaying the change process from 0 to 200.
		405.Zr	置零范围 Zero range	500	0-50000	零点绝对值在此参数范围内才允许手动置零、自动置零 The absolute value of zero point can only be manually or automatically reset within this parameter range
		406.PZ	上电置零开关 Power on zero switch	0	0-1	0: close 1: Open
		407.Pt	上电置零时间 Power on reset time	10	0-1800	重新上电会按此时间倒计时 Re powering on will count down to this time
		408.Pr	上电置零范围 Power on zero range	50	0-50000	上电置零开启时，重新上电，检测到零点绝对值在此参数范围内时，零点自动置零 When the power is turned on and reset to zero, when the power is turned on again and the absolute value of the zero point is detected

						within this parameter range, the zero point will be automatically reset to zero
		409.AZ	自动置零开关 Automatic zero switch	0	0-1	0: close 1: Open
		410.At	自动置零时间 Automatic zeroing time	0.10	0.01-9.999	
		411.Ar	自动置零范围 (分度) Automatic Zero Range (Divisions)	1.0	0.1-50.0	在自动置零时间内，零点绝对值在自动置零范围内时，自动置零（此参数 X 分度值为最终的范围，区别于【401 显示归零范围】，自动置零后零点为 0） *【412】【413】判定“重量稳定”时生效 During the automatic zeroing time, when the absolute value of the zero point is within the automatic zeroing range, it is automatically zeroed (this parameter's X division value is the final range, which is different from [401 Display Zero Range], where the zero point is 0 after automatic zeroing) *Effective when determining "weight stability"
		412.Wr	判稳范围 (分度) Determine stable range (Graduation)	0	0-5	判定“重量稳定”是否生效， =0: 代表不判稳，重量始终稳定 >0: 【413】判稳时间内，重量变化处于【412】判稳范围内时，表示“重量稳定” 此参数影响手动去皮，自动置零 此参数 X 分度值为最终的范围 Determine whether "weight stability" is effective, =0: represents instability, weight is always stable > 0: When the weight change is within the stable range of 【 412 】 within the stable time of 【 413 】 , it indicates "weight stability"

C5.CoM	485 通讯 485 communi cation					This parameter affects manual peeling and automatic zeroing. The X division value of this parameter is the final range
		413.Wt	判稳时间 Determine stable time	1.000	0.01-9.999	【412】【413】两个参数共同判定是否为稳定状态 【 412 】 【 413 】 Two parameters jointly determine whether it is in a stable state
		500.Ar	通讯地址 Mailing address	1	1-253	
		501.br	波特率 Baud rate	3	1-15	1:2400 2:4800 3:9600 4:19200 5:22800 6:38400 7:57600 8:115200 9:128000 10:230400 11:256000 12:460800 13:500000 14:512000 15:600000 修改波特率立即生效，需断电重新连接，由于显示位数有限，显示时去掉最后面两位，如 2400 显示 24，15200 显示 1152 The modification of the baud rate takes effect immediately and requires power outage and reconnection. Due to limited display digits, the last two digits are removed during display, such as 2400 displaying 24 15200 displaying 1152
		502.Vb	校验位 Check bit	0	0-2	0: none（无校验） 1: even（奇校验） 2: odd（偶校验） 0: none (no verification) 1: Even (odd check) 2: Odd (Even Check)
		503.so	停止位 Stop bit	1	1/2	停止位 1 位或 2 位 Stop bit 1 or 2
		504.AS	主动发送模式 Active sending mode	0	0-1	0: close 1: Open 主动发送模式为关闭时表示 modbus-rtu 通讯模式 When the active sending mode is turned off, it indicates modbus rtu communication mode

		505.AF	主动发送频率 Active transmission frequency	2	0-9	0:1Hz 1:2Hz 2:5Hz 3:10Hz 4:20Hz 5:25Hz 6:60Hz 7:100Hz 8:500Hz 9:1000Hz	
C6.aNa	模拟量输出 Analog quantity output	600.At	AO 输出类型 AO output type	1	0-3	0:NONE (无) 1:GroSS (总重) 2:NEt (净重) 3:pEak (峰值) 0: NONE 1: GroSS 2: NEt 3: pEak	
		601.As	AO 信号类型 AO signal type	0	0-1	0:Curr (电流) 1:VoLt (电压) 0: Curr 1: VoLt	
		602.ax	AO 输出最大重量 AO output Maximum weight	5000	0-99999	模拟量输出对应的满量程 Full range corresponding to analog output	可双极性输出, 当【604】选择极性为负时, AO 输出的最小重量为负向的满量程
		603.ai	AO 输出最小重量 AO output minimum weight	0	0-9999	模拟量输出对应的零点 Zero point corresponding to analog output	Capable of bipolar output, when 【 604 】 selects negative polarity, the minimum weight of AO output is full scale in the negative direction
		604.aP	AO 输出最小重量极性 AO output minimum weight polarity	0	0-1	0:plus (正) 1:miNus (负) 此参数选择 1 时, 可以实现 4-12-20mA, 0-5-10V 输出 0: plus (positive) 1: miNus (negative) When this parameter is selected as 1, it can achieve outputs of 4-12-20mA and 0-5-10V	
		605.CL	电流输出下限 Lower limit of current output	4	0-10.00	0-20mA 输入 0 4-20mA,4-12-20mA 输入 4 0-20mA input 0 4-20mA, 4-12-20mA input 4 0-20mA input 0 4-20mA, 4-12-20mA input 4 0-20mA input 0	

						4-20mA, 4-12-20mA input 4
		606.Ch	电流输出上限 Current output upper limit	20	10.00-21.00	
		607.VL	电压输出下限 Lower limit of voltage output	0	0-5.00	
		608.Vh	电压输出上限 Voltage output upper limit	10.00	5.00-10.00	0-5V 输入 5.00 0-10V, 0-5-10V 输入 10.00 0-5V input 5.00 0-10V, 0-5-10V input 10.00
C7.LOC	登陆密码 Login password	700.oP	登陆密码开关 Login password switch	0	0-1	0: close 1: Open
		701.PW	修改密码 Change password	0000	0-9999	默认密码 0000, 修改时需要输入默认密码才能更改新密码 The default password is 0000. To change it, you need to enter the default password in order to change the new password
C8.FAC	出厂调校 Factory calibration	800.Lo	厂家密码 Manufacturer password			厂家内部使用 Internal use by the manufacturer
		801.EX	退出出厂调校 Exit factory calibration			
		802.01	AO 0.1mA 校准 AO 0.1mA calibration			
		803.4m	AO 4mA 校准 AO 4mA calibration			
		804.10	AO 10mA 校准 AO 10mA calibration			
		805.12	AO 12mA 校准 AO 12mA calibration			
		806.20	AO 20mA 校准 AO 20mA calibration			
		807.UU	AO 10V 校准 AO 10V calibration			

		808.Ad	ADC 基准设置 ADC benchmark setting			
C9.iNF	版本信息 Version information	900.VE	软件版本号 Software version number			
		901.id	设备唯一编码 Unique device code			

故障或字符说明 Fault or character description

副屏显示 Secondary screen display	故障或字符说明 Fault or character description
PoiNt	上电初始化状态，开机时显示，在获取到第一个有效数据后自动退出，如果打开了上电置零功能，在倒计时结束时退出 Power on initialization status, displayed on startup, automatically exits after obtaining the first valid data. If the power on reset function is enabled, exits at the end of the countdown
-----	没有捕捉到任何有效峰值 No effective peak was captured
Err.ro	故障 EEPROM 读写失败了，用户按确定键解除 Fault EEPROM read/write failed, user presses OK key to resolve
Err.Ad	ADC 故障 ADC malfunction
AL.Ad	ADC 故障，采样信号过大，ADC 读到了最大或者最小值了（自动解除） ADC malfunction, sampling signal too large, ADC reading maximum or minimum value (automatic release)
AL.oL	传感器过载或超过最大量程 Sensor overload or exceeding maximum range
AL.AZR	传感器上电清零失败，超过设置的上电清零值了 Sensor power on reset failed, exceeding the set power on reset value
AL.PEL	手动去皮失败，不满足去皮条件 Manual peeling failed, not meeting the peeling conditions
AL.MZr	手动置零失败，不满足置零条件 Manual zeroing failed, not meeting the zeroing conditions
主屏显示 Main screen display	故障或字符说明 Fault or character description
-----	修改采样速率时出现，自动退出 Appears when modifying sampling rate, automatically exits
Er.doM	用户设置的值太小了，会自动设置成可以设置的最小值 The value set by the user is too small and will be automatically set to the minimum value that can be set
Er.up	用户设置的值太大了，会自动设置成可以设置的最大值 The value set by the user is too large and will be automatically set to the maximum value that can be set
Er.iNV	用户输入无效 User input is invalid
Er.pwE	用户输入密码错误 User entered password incorrectly

FA.roM	配置的参数写入失败，但参数已经生效 The configuration parameters failed to be written, but the parameters have already taken effect
FA.AdC	ADC 设置失败 ADC setting failed
Er.buy	系统忙无法执行操作，一般是数据没有采集稳定就执行了清零或校准操作了 The system is busy and unable to perform operations, usually due to data not being collected stably before performing zeroing or calibration operations

主从式 modbus-RTU 通信寄存器表
Master slave modbus RTU communication register table

数据名 Data Name	地址 address			有效值类型 Valid value types	读写 Reading and writing	功能说明 Function Description
	10 进制 10 base	16 进制 Hexadecimal	PLC			
总重 Total weight	0	0X0000	40001	32 位有符号 32-bit signed	低 low R	
	1	0X0001	40002		高 high R	
净重 Net weight	2	0X0002	40003	32 位有符号	低 low R	
	3	0X0003	40004		高 high R	
峰值 peak value	4	0X0004	40005	32 位有符号	低 low R	
	5	0X0005	40006		高 high R	
内码值 Internal code value	6	0X0006	40007	32 位有符号	低 low R	
	7	0X0007	40008		高 high R	
小数点 decimal point	8	0X0008	40009	16 位无符号 16 bit unsigned	R/W	0:00000 1:0000.0 2:000.00 3:00.000 4:0.0000
单位 unit	9	0X0009	40010	16 位无符号	R/W	0: NoNE (none) 1: G (g) 2: kG (kg) 3: t (T) 4: N (N) 5: pa (pa) 6: kp (Kpa) 7: mpa (Mpa) 8: N.M (N.m) 9: kN (kN)
状态 state	10	0X000A	40011	16 位无符号	R	0: 数据有效状态 1: 峰值检测状态 2: ROM 故障 3: ADC 故障 4: ADC 信号过大 5: 总重超量程 (过载)

						6: 上电置零失败 7: 不满足去皮条件 8: 超出置零范围 9: 继电器 1 状态 10: 继电器 2 状态 11: 继电器 3 状态 0: Data validity status 1: Peak detection status 2: ROM malfunction 3: ADC malfunction 4: ADC signal too large 5: Total weight exceeding range (overload) 6: Power on zero setting failure 7: Not meeting the peeling conditions 8: Beyond the zero setting range 9: Relay 1 status 10: Relay 2 status 11: Relay 3 status	
命令 command	11	0X000B	40012	16 位无符号	R/W	1: 去皮（不保存） 2: 去皮（保存） 3: 取消去皮（保存） 4: 置零（不保存） 5: 校零(保存) 6: 清除峰值 7: 校准 9: 恢复出厂 1: Peeling (not stored) 2: Peeling (preservation) 3: Cancel Peeling (Save) 4: Zero (not saved) 5: Zero calibration (save) 6: Clear Peak 7: Calibration 9: Restore Factory	
砝码值 Weight value	12	0X000C	40013	32 位有符号	低	R/W	校准时输入的砝码重量
	13	0X000D	40014		高	R/W	The weight of the weight input during calibration
蠕变追踪 Creep tracking	14	0X000E	40015	16 位无符号	R/W	0-10	
显示归零 Display zeroing	15	0X000F	40016	16 位无符号	R/W	0-5000	
动态追踪范围	16	0X0010	40017	16 位无符号	R/W	0-5000	

Dynamic tracking range						
动态追踪更新时间 Dynamic tracking update time		17	0X0011	40018	16 位无符号	R/W 0-2000 (单位 ms)
稳定重量开关 Stable weight switch		18	0X0012	40019	16 位无符号	R/W 0:关 1: 开
置零范围 Zero range		19	0X0013	40020	16 位无符号	R/W 0-5000
上电置零使能 Power on zero enable		20	0X0014	40021	16 位无符号	R/W 0:关 1: 开
上电置零时间 Power on reset time		21	0X0015	40022	16 位无符号	R/W 0-1800 (单位 S)
上电置零范围 Power on zero range		22	0X0016	40023	16 位无符号	R/W 0-50000
自动置零使能 Automatic Zero Enable		23	0X0017	40024	16 位无符号	R/W 0:关 1: 开
自动置零时间 Automatic zeroing time		24	0X0018	40025	16 位无符号	R/W 100-9999 (单位 ms)
自动置零范围 Automatic Zero Range		25	0X0019	40026	16 位无符号	R/W 10-500 (单位 0.1)
判稳范围分度 Stable range division		26	0X001A	40027	16 位无符号	R/W 0-5
判稳时间 Determine stable time		27	0X001B	40028	16 位无符号	R/W 100-9999 (单位 ms)
继电器 1	工作模式 Working mode	28	0X001C	40029	16 位无符号	R/W 0: 不输出 1: 上限动作 2: 下限动作 3: 区间内动作 4: 区间外动作 0: No output 1: Upper limit action 2: Lower limit action 3: Action within the interval 4: Action outside the interval

	回差 Retracement	29	0X001D	40030	16 位无符号		R/W	
	数据类型 data type	30	0X001E	40031	16 位无符号		R/W	
	动作延迟 Action delay	31	0X001F	40032	16 位无符号		R/W	
	输出上限 Output upper limit	32	0X0020	40033	16 位无 符号	低	R/W	
		33	0X0021	40034		高	R/W	
	输出下限 Output Lower Limit	34	0X0022	40035	16 位无 符号	低	R/W	
		35	0X0023	40036		高	R/W	
	工作模式 Working mode	36	0X0024	40037	16 位无符号		R/W	同继电器 1 Same as relay 1
	动作回差 Action return difference	37	0X0025	40038	16 位无符号		R/W	
继 电 器 2	数据类型 data type	38	0X0026	40039	16 位无符号		R/W	
	动作延迟 Action delay	39	0X0027	40040	16 位无符号		R/W	
	输出上限 Output upper limit	40	0X0028	40041	16 位无 符号	低	R/W	
		41	0X0029	40042		高	R/W	
	输出下限 Output Lower Limit	42	0X002A	40043	16 位无 符号	低	R/W	
		43	0X002B	40044		高	R/W	
	工作模式 Working mode	44	0X002C	40045	16 位无符号		R/W	同继电器 1 Same as relay 1
继 电 器 3	动作回差 Action return difference	45	0X002D	40046	16 位无符号		R/W	
	数据类型 data type	46	0X002E	40047	16 位无符号		R/W	
	动作延迟 Action delay	47	0X002F	40048	16 位无符号		R/W	
	输出上限 Output upper limit	48	0X0030	40049	16 位无 符号	低	R/W	
		49	0X0031	40050		高	R/W	
	输出下限 Output Lower Limit	50	0X0032	40051	16 位无 符号	低	R/W	
		51	0X0033	40052		高	R/W	

